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(kali㉿kali)-[~/vaje15]
$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 5 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 5u
gpg: next trustdb check due at 2026-12-17
/home/kali/.gnupg/pubring.kbx
-----
pub   rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
      9EEDFB1542D4D8D0916EB1A18DF495E7B56EC5CD
uid           [ultimate] Zans Test (KV-Vaje) <zans@kv.test>
sub   rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub   rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
      2C88FDFF21C89B21CD491009E9AA2BDF0968E3AB
uid           [ultimate] Zans2 Test (Vaje2) <zankv2@kv.test>
sub   rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub   rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
      C7C9F95E071519793B33CF32BD124003A7D6E5CE
uid           [ultimate] Alice (AliceVaje) <alice@example.com>
sub   rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub   rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
      B2202CAE05E9113953D5CF552A4F5E6308C00E1C
uid           [ultimate] Bob (BobVaje) <bob@example.com>
sub   rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub   rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
      01FCDA6A9324E4E94EEB86D6A04B16DB60A9D046
uid           [ultimate] Mallory (MalloryVaje) <mallory@example.com>
sub   rsa4096 2025-12-17 [E] [expires: 2026-12-17]

(kali㉿kali)-[~/vaje15]
$ █

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(kali㉿kali)-[~/vaje15]
$ gpg --armor --export mallory@example.com > bob_pubkey.asc

(kali㉿kali)-[~/vaje15]
$ cat bob_pubkey.asc
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQINBGLC2GEBEADUPEyJSDyauvU50nnRi4BE2l9bnbRQXH89nEdf2qreV1h7Gvj4
lLXOYuHcAYiFNoYZKXQ9bGgAZAAXEF/KDq8mEDUf2USlhv0loaIT1PDuaFo0RtbU
pBVCC1QXpUE6X0asbDEoiZTWMT91YXywzsLcUae1wjwpxyE7yKMoo5G9WcdXOBPr
2LiB8rVJY7h2yxv/r6tt7XWCiDl8PRXk/qiDSYWwFeq7mwZsc6joxNZky/XyxpVv
PZjpZRGvGkYoryCMZr5HvpF7aPn3EKL0NyJWGOh9tpgH4msS5Czecd+C4kjGMZYG
eI7hWHBDCInbz/Tp4oB/i6iwe0v76b540HsL8lNvxUWggXc3nxMpl0DjYioJxn3t
PKj1jEeVz90//Am2JD3Quw0H6/pCtoanDeRLCogwYeEAYnDWLqJ9RTVh6X4xl0sY
hnpU15QoSXaUhrAjqgIdewcCvvbTEnzLVA/3hGbG9qVoCgHzipjiM/fIRkvXEA7g
xV87ANw8tYF9MgrU4dE2AzQiTwfuIa6KKR3CxEPyB10jz7Hexp0SdKyViZLRym1r
prSu43ztC1q2cRV/8N3U0Qn7N+40jSc5vHyRUxDAgefoHDCAS93kQy5gErYJjSks
q+UkNwGeABHF3hCtM5Jqt9W0mpaD6e+dbBgII4hZgZRvXVEwqGjR+z+XKQARAQAB
-----

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(kali㉿kali)-[~/vaje15]
$ echo "Zaupno sporocilo za Boba" > secret.txt

(kali㉿kali)-[~/vaje15]
$ cat secret.txt
Zaupno sporocilo za Boba

(kali㉿kali)-[~/vaje15]
$ gpg --encrypt --recipient bob@example.com secret.txt

(kali㉿kali)-[~/vaje15]
$ cat secret.txt.gpg
+ma`+p@FRX5^+
=U+s*QV*64+H+^+{+>+d+g+I+...+K+s*M+y*8*R+BZhH)$v+v+M+K,c+y+Q+6+ 形+se3+.+f%+f+f+|+D+'b
i-b+++ac+++?+U+CG*1+7y|+P+er)+$+++rQ M+)+8+8 V+++i ++>Q+l +vG+++x+^+o:-Q++$++f+g[+T+w$+++Zl++(B`@D+=+S+m++++T3$8 +<+e+++}Y+u
+84e+|++C+c+q+J++++bU+H+Z++0+++vj+^aVen++0*[U+@+',G+6+V+++/*}N++B>+o/++++}+++++8+++rS+++++I+4{V+++W+e+>v++]+ ++h++}wV+er +k+
+z+++r++4++X+0+q+++J+0++//+uKQ +9ya+0++s+++/"*^J+++g+*8+Q]Q>11+)+Q
R+*\+B*x

(kali㉿kali)-[~/vaje15]
$ gpg --decrypt secret.txt.gpg
gpg: encrypted with rsa4096 key, ID B2AB6D61196016C3, created 2025-12-17
"Bob (BobVaje) <bob@example.com>"
Zaupno sporocilo za Boba

(kali㉿kali)-[~/vaje15]
$

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(kali㉿kali)-[~/vaje15]
$ gpg --edit-key bob@example.com
gpg (GnuPG) 2.4.8; Copyright (C) 2025 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Secret key is available.

sec  rsa4096/2A4F5E6308C00E1C
     created: 2025-12-17  expires: 2026-12-17  usage: SC
     trust: ultimate      validity: ultimate
ssb  rsa4096/B2AB6D61196016C3
     created: 2025-12-17  expires: 2026-12-17  usage: E
[ultimate] (1). Bob (BobVaje) <bob@example.com>

gpg> trust
sec  rsa4096/2A4F5E6308C00E1C
     created: 2025-12-17  expires: 2026-12-17  usage: SC
     trust: ultimate      validity: ultimate
ssb  rsa4096/B2AB6D61196016C3
     created: 2025-12-17  expires: 2026-12-17  usage: E
[ultimate] (1). Bob (BobVaje) <bob@example.com>

Please decide how far you trust this user to correctly verify other users' keys
(by looking at passports, checking fingerprints from different sources, etc.)

  1 = I don't know or won't say
  2 = I do NOT trust
  3 = I trust marginally
  4 = I trust fully
  5 = I trust ultimately
  m = back to the main menu

Your decision? 5
Do you really want to set this key to ultimate trust? (y/N) y

sec  rsa4096/2A4F5E6308C00E1C
     created: 2025-12-17  expires: 2026-12-17  usage: SC
     trust: ultimate      validity: ultimate
ssb  rsa4096/B2AB6D61196016C3
     created: 2025-12-17  expires: 2026-12-17  usage: E
[ultimate] (1). Bob (BobVaje) <bob@example.com>

gpg> quit

(kali㉿kali)-[~/vaje15]
$

```

```

(kali@kali)-[~/vaje15]
$ gpg --edit-key mallory@example.com
gpg (GnuPG) 2.4.8; Copyright (C) 2025 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Secret key is available.

sec  rsa4096/A04B16DB60A9D046
    created: 2025-12-17  expires: 2026-12-17  usage: SC
    trust: ultimate      validity: ultimate
ssb  rsa4096/DC320007C3FB5EB3
    created: 2025-12-17  expires: 2026-12-17  usage: E
[ultimate] (1). Mallory (MalloryVaje) <mallory@example.com>

gpg> trust
sec  rsa4096/A04B16DB60A9D046
    created: 2025-12-17  expires: 2026-12-17  usage: SC
    trust: ultimate      validity: ultimate
ssb  rsa4096/DC320007C3FB5EB3
    created: 2025-12-17  expires: 2026-12-17  usage: E
[ultimate] (1). Mallory (MalloryVaje) <mallory@example.com>

Please decide how far you trust this user to correctly verify other users' keys
(by looking at passports, checking fingerprints from different sources, etc.)

  1 = I don't know or won't say
  2 = I do NOT trust
  3 = I trust marginally
  4 = I trust fully
  5 = I trust ultimately
  m = back to the main menu

Your decision? 2

sec  rsa4096/A04B16DB60A9D046
    created: 2025-12-17  expires: 2026-12-17  usage: SC
    trust: never         validity: ultimate
ssb  rsa4096/DC320007C3FB5EB3
    created: 2025-12-17  expires: 2026-12-17  usage: E
[ultimate] (1). Mallory (MalloryVaje) <mallory@example.com>
Please note that the shown key validity is not necessarily correct
unless you restart the program.

gpg> quit

```

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(kali@kali)-[~/vaje15]
$ gpg --fingerprint
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid: 4  signed: 0  trust: 0-, 0q, 0n, 0m, 0f, 4u
gpg: next trustdb check due at 2026-12-17
/home/kali/.gnupg/pubring.kbx

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
    9EED FB15 42D4 D8D0 916E B1A1 8DF4 95E7 B56E C5CD
uid      [ultimate] Zans Test (KV-Vaje) <zans@kv.test>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
    2C88 FDFD 21C8 9B21 CD49 1009 E9AA 2BDF 0968 E3AB
uid      [ultimate] Zans2 Test (Vaje2) <zankv2@kv.test>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
    C7C9 F95E 0715 1979 3B33 CF32 BD12 4003 A7D6 E5CE
uid      [ultimate] Alice (AliceVaje) <alice@example.com>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
    B220 2CAE 05E9 1139 53D5 CF55 2A4F 5E63 08C0 0E1C
uid      [ultimate] Bob (BobVaje) <bob@example.com>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
    01FC DA6A 9324 E4E9 4EEB 86D6 A04B 16DB 60A9 D046
uid      [ unknown] Mallory (MalloryVaje) <mallory@example.com>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

```

Razmislek:

1. GPG ne more vedeti ali je javni ključ od prave osebe. Preveri le kriptografsko pravilnost ključa.
2. Je hash javnega ključa za njegovo identiteto. Omogoči preverjanje če je nek javni ključ pravilen.
3. Zato ker se lahko v epošto dokaj enostavno prestreže. Napadalec lahko zamenja ključ v epošti in se izdaja za drugo osebo. Epošta tudi ni end to end šifrirana.
4. Temelji na medsebojnem preverjanju in zaupanju med uporabniki. Ko več oseb podpiše javni ključ, se večja verjetnost resničnosti.